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SMART DESK:An Intelligent System for Automated IT Support Ticket Analysis, NLP Classification, Team Routing, Resolution Time Prediction, and Anomaly Detection

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1. Executive Summary

IT support teams receive large volumes of tickets that must be classified, routed to the right team, and resolved quickly to avoid SLA breaches. This project builds a data-driven pipeline that processes 100,000 historical IT support tickets to (1) classify the issue type from the customer's message using NLP, (2) route each ticket to the responsible team, (3) estimate resolution time, and (4) detect abnormal days in ticket activity.

- **Issue type classification:** TF-IDF + Random Forest reached 100% accuracy. A grouped-split sanity check showed the data is built from only 96 message templates, and on unseen messages the Macro F1 was 0.605 ± 0.107 (5-fold GroupKFold).
- **Team routing:** Logistic Regression on the message text reached 57.4% accuracy (vs. a 14.4% baseline) and 71.6% top-3 accuracy. On unseen messages, the Macro F1 was 0.516 ± 0.112 (5-fold GroupKFold).
- **BERT vs TF-IDF:** Pre-trained Sentence-BERT embeddings performed on par with TF-IDF, so the proposal's hypothesis that BERT would clearly outperform keyword methods was not supported on this templated dataset.
- **Resolution time:** After removing data leakage, eight models all converged to the median baseline (MAE ≈ 15.2 hours). Mutual information confirmed the features carry almost no signal.
- **Anomaly detection:** Isolation Forest and LOF agreed on 45 high-confidence anomalous days out of 1,460, and the detector caught 10 out of 10 artificially injected surges.
- **Dashboard:** An interactive Streamlit dashboard brings the results together for helpdesk managers, including live ticket classification and routing.

2. Introduction & Problem Statement

Manual sorting and routing of thousands of IT support tickets costs companies valuable time, delays employee workflows, and leads to SLA (Service Level Agreement) breaches. Automating helpdesk intelligence bridges the gap between raw user requests and fast, precise technical interventions.

Business question: How can organizations automatically process and analyze large volumes of IT support tickets using NLP to classify issues, route tickets to the correct teams, predict resolution times, and detect anomalous ticket surges before operational bottlenecks disrupt business continuity?

3. Objectives & Scope

- Clean and analyze historical ticket data to understand support patterns.
- Classify ticket issue types automatically from the customer's message using NLP.
- Route tickets to the responsible team (product area).
- Compare BERT embeddings with TF-IDF text representations.
- Predict ticket resolution time.
- Detect unusual daily patterns in ticket volume, resolution time, and reopen rate.
- Present insights through an interactive Streamlit dashboard.

Scope change: Priority assignment was removed from the original proposal because the dataset does not contain priority labels.

4. Data Description

Source: Synthetic IT support tickets dataset (`synthetic_it_support_tickets.csv`).

Size: 100,000 tickets × 19 columns, covering 1 January 2022 to 30 December 2025 (1,460 days).

Table 4.1: Dataset columns

Column type	Columns
Identifiers	ticket_id, customer_id
Time	created_at
Categorical	customer_segment, channel, product_area, issue_type, status, sla_plan, customer_sentiment, platform, region
Text	initial_message, agent_first_reply, resolution_summary
Numerical	resolution_time_hours, reopened, csat_score, has_attachment

Targets: `issue_type` (8 classes) for issue classification, `product_area` (7 classes) for team routing, and `resolution_time_hours` for regression.

5. Data Cleaning & Preprocessing

Table 5.1: Data quality checks

Check	Result / Action
Duplicate ticket IDs	0 found
Negative resolution times	0 found
CSAT outside 1–5 range (excluding 0)	0 found
Invalid status values	0 found
CSAT = 0 (29,941 rows, 29.9%)	Means “not rated”; converted to missing (NaN) so it is excluded from CSAT averages
Missing region (19,997 rows)	Filled with "Unknown"
Missing resolution_summary (39,887 rows)	Filled with "Pending"
Missing resolution_time_hours (39,887 rows)	Kept as missing: these tickets are open, in progress, or on hold

Outlier analysis: Using the IQR method, 6,108 resolution times exceeded the upper bound of 106.83 hours. These values were kept because they are genuine, mostly belonging to tickets closed without action (see Section 6).

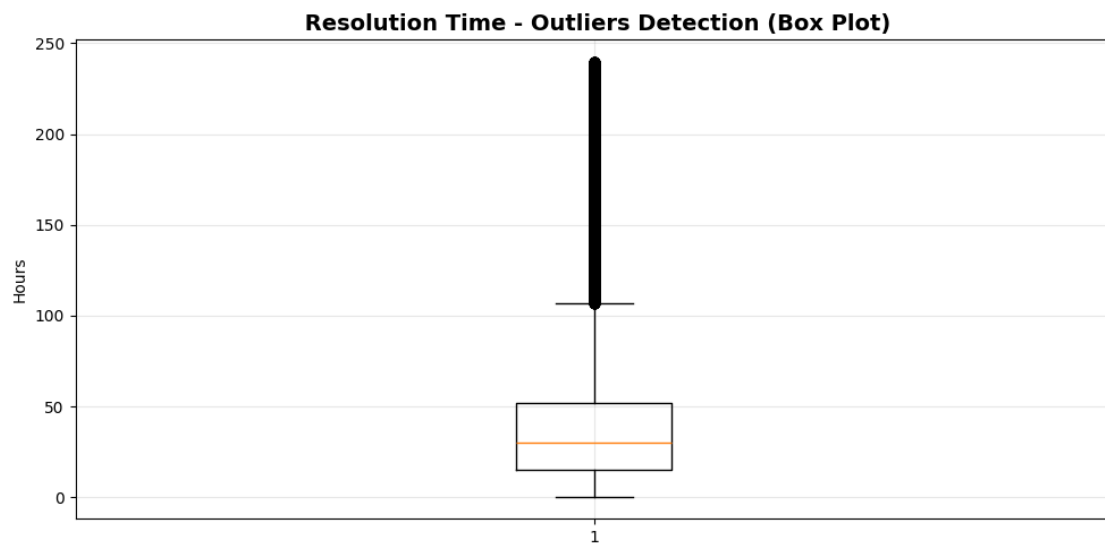


Figure 5.1: Resolution time box plot

6. Exploratory Data Analysis (EDA)

6.1 Ticket Distributions

- Issue types are balanced (12,349–12,739 tickets each), and product areas are balanced (14,099–14,432 each), so no class balancing was required.
- Tickets are evenly spread across hours of the day and days of the week, with no clear peak submission times.
- Customer sentiment: neutral (33%), negative (26%), very negative (16%), positive (16%), very positive (9%).
- About 5% of tickets were reopened, with similar rates across issue types (4.8%–5.3%).

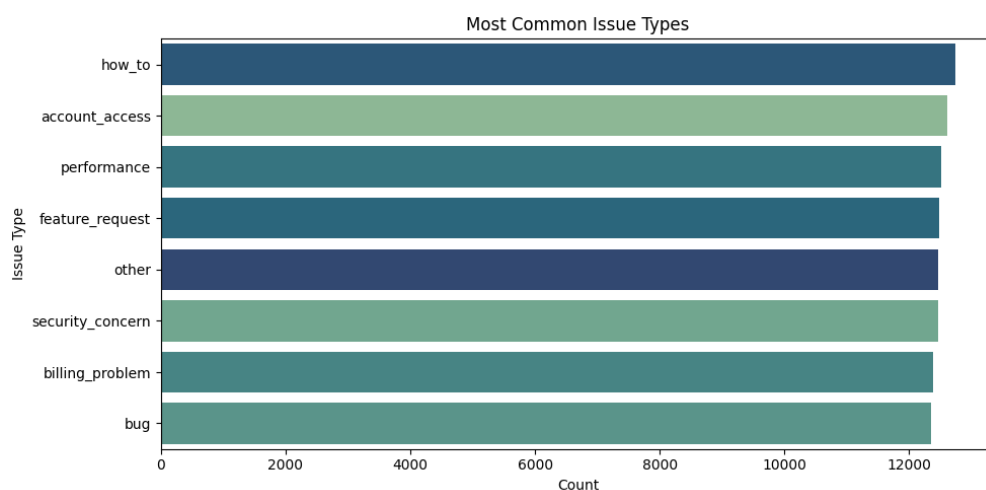


Figure 6.1: Most common issue types

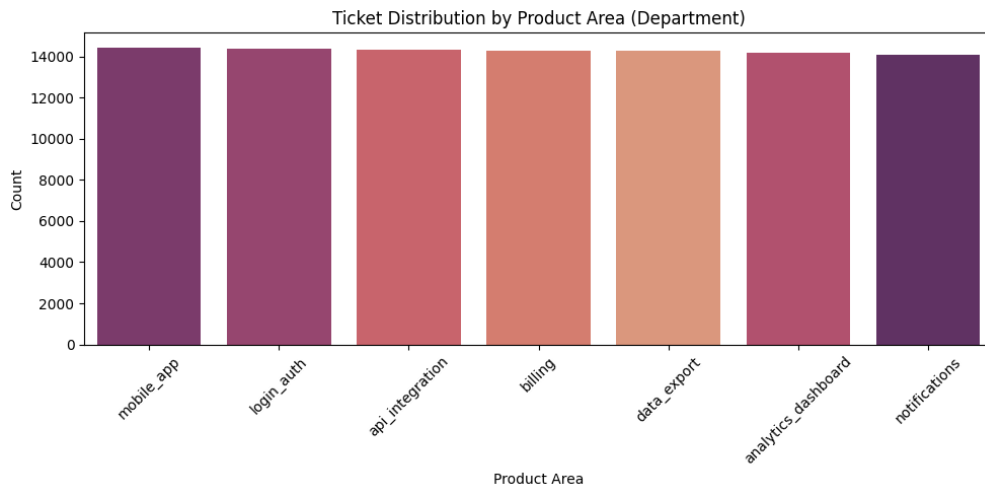


Figure 6.2: Ticket distribution by product area

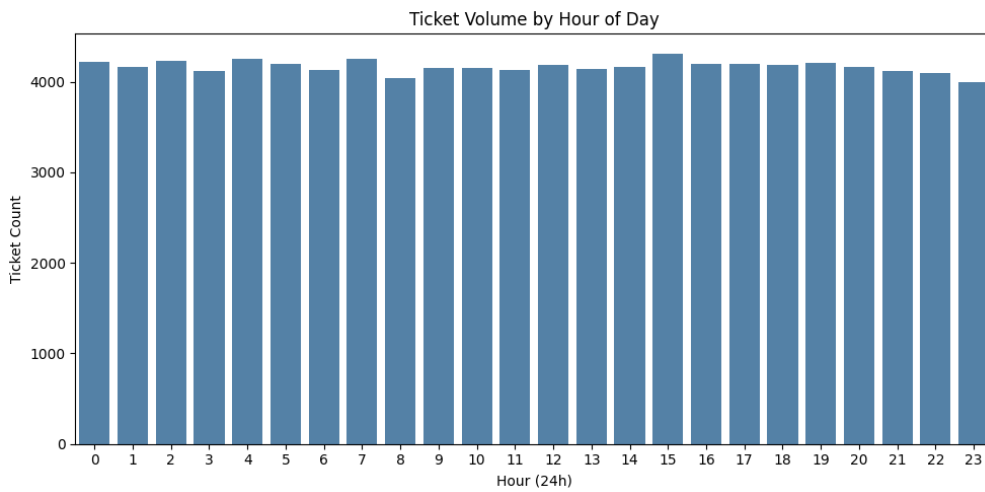


Figure 6.3: Ticket volume by hour of day

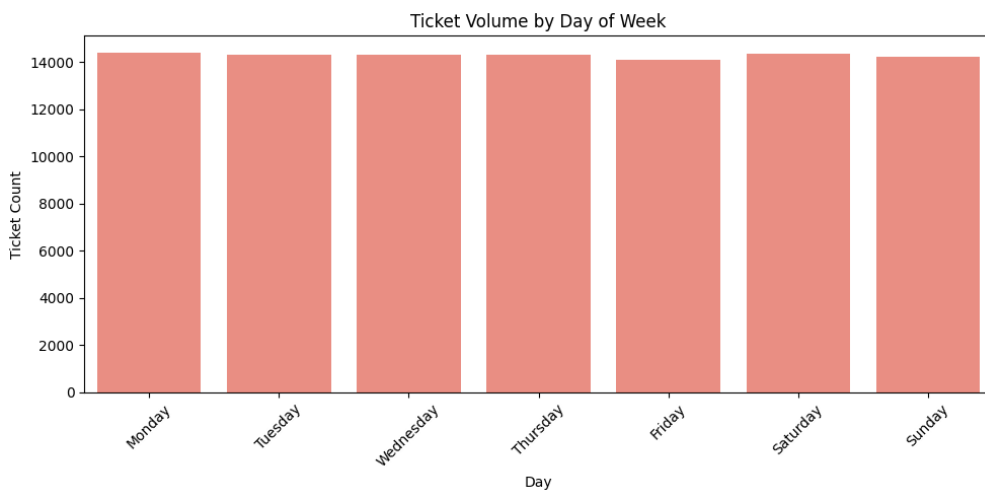


Figure 6.4: Ticket volume by day of week



Figure 6.5: Customer sentiment distribution

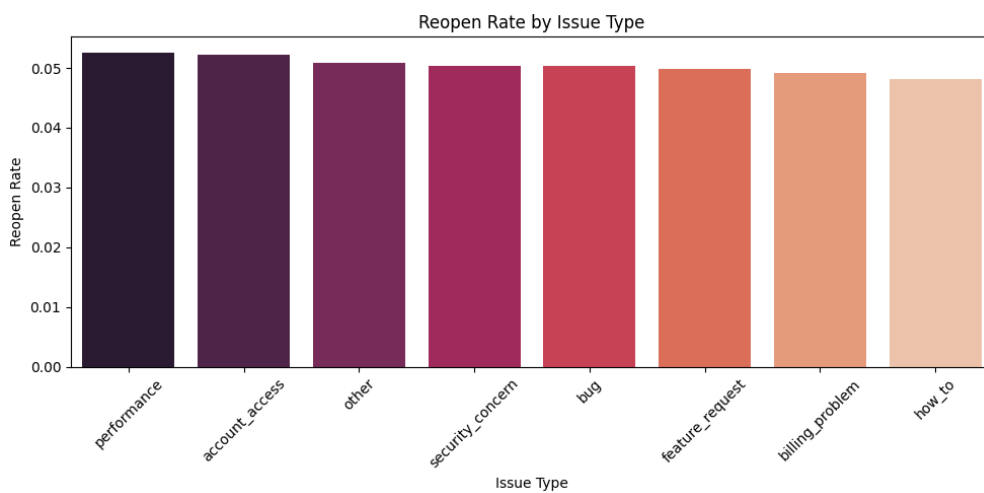


Figure 6.6: Reopen rate by issue type

6.2 Resolution Time

- Median resolution time is almost identical (~30 hours for closed tickets) across issue types, customer segments, and regions.
- The only strong difference is by status: resolved tickets have a median of ~23.5 hours, while tickets closed without action have a median of ~131 hours.
- Average CSAT is nearly identical across regions.

Key insight: Most ticket attributes show no relationship with resolution time. This finding directly explains the resolution time modeling results in Section 10.

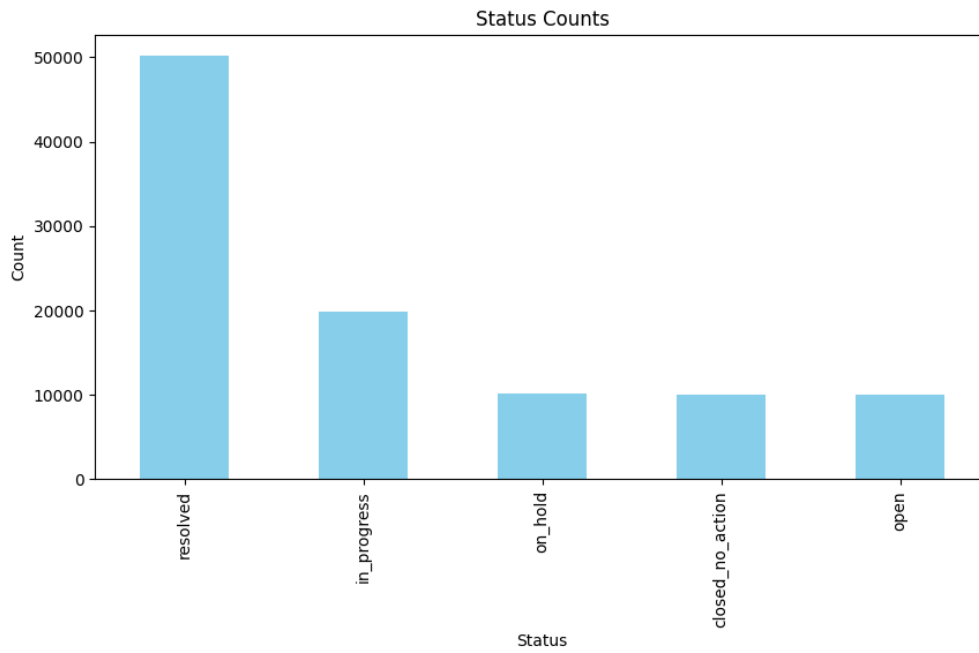


Figure 6.7: Ticket count by status

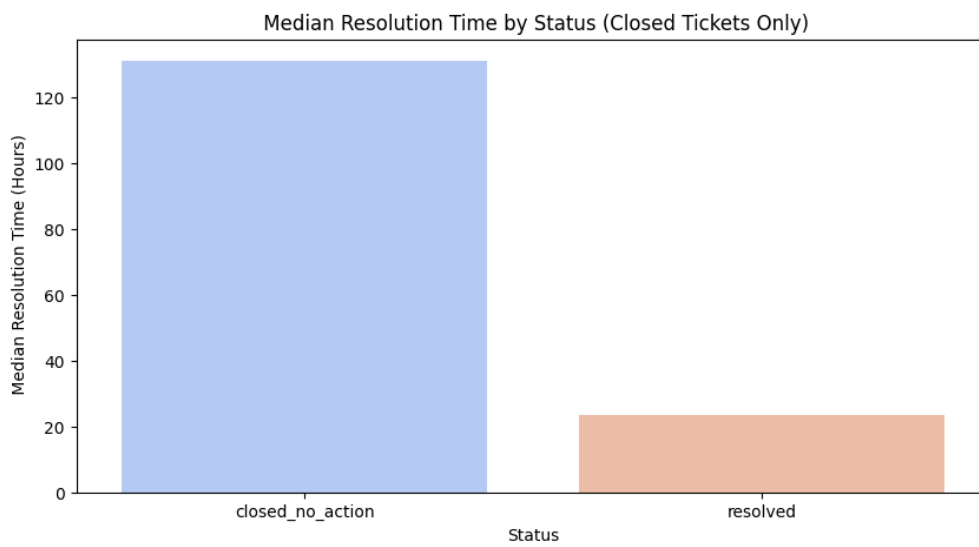


Figure 6.8: Median resolution time by status (closed tickets)

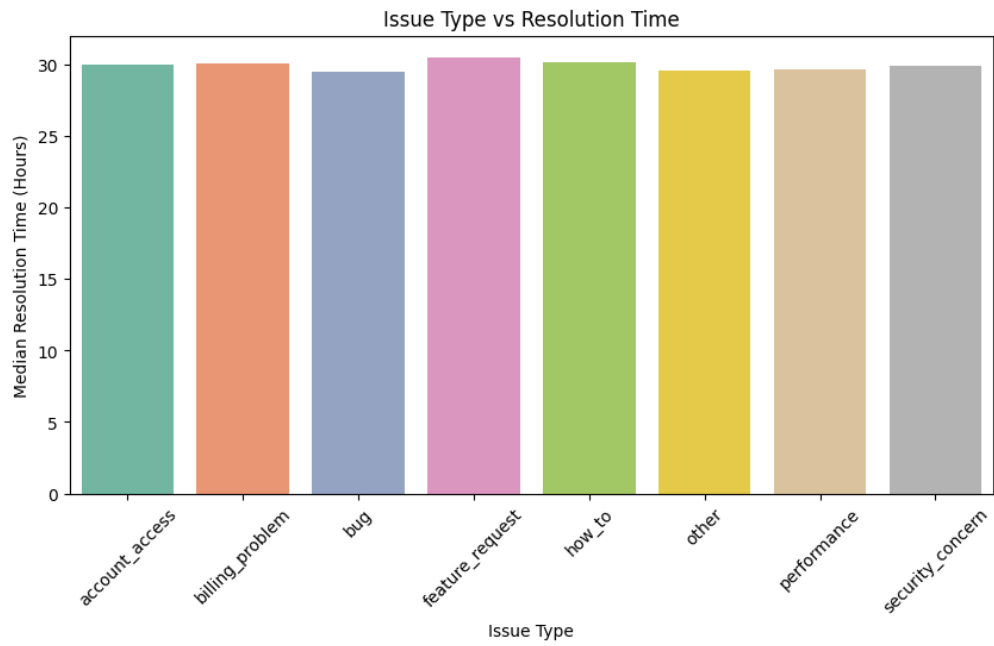


Figure 6.9: Median resolution time by issue type

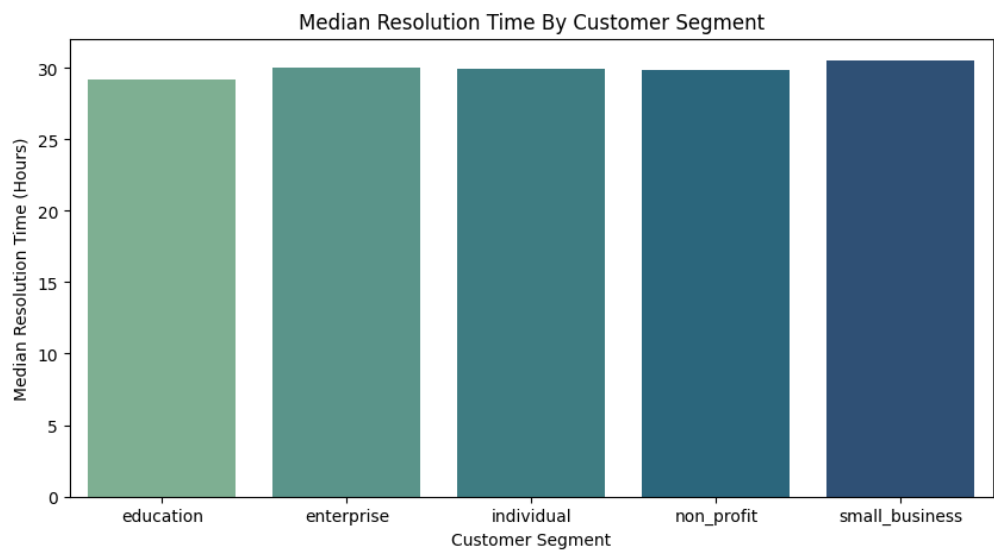


Figure 6.10: Median resolution time by customer segment

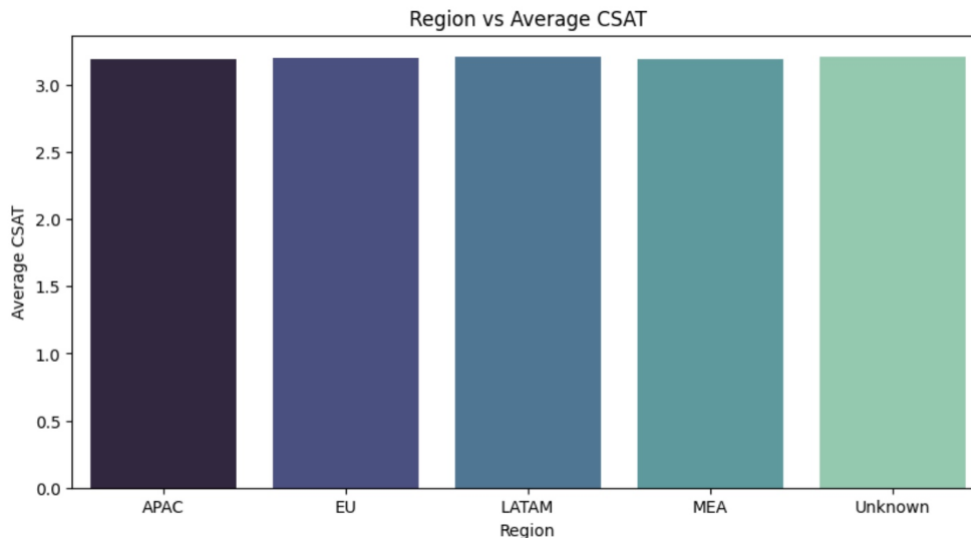


Figure 6.11: Average CSAT by region

7. Feature Engineering & Text Preprocessing

7.1 Structured Features

- **Time features:** hour of creation and day of week extracted from `created_at`.
- **Message features:** message length, word count, and presence of question marks and exclamation marks.
- **Customer feature:** number of tickets per customer.
- **Encoding:** One-Hot Encoding for categorical columns (51 columns after encoding); Label Encoding for classification targets.
- **Dropped columns:** identifiers and raw text columns not used as direct model input.

Leakage prevention: Features only known after a ticket is closed were removed from the resolution time model: `status`, `csat_score`, `reopened`, and CSAT-based aggregates (`customer_avg_csat`, `region_avg_csat`). For routing, only fields available when a ticket is opened were used (message, channel, platform, customer segment).

7.2 Text Preprocessing (NLP)

The customer's `initial_message` was processed with NLTK: lowercasing, tokenization, stop-word removal (keeping meaningful words such as "not", "no", and "how"), and lemmatization. For example, "I cannot log in; the system says my password is incorrect" became "cannot log system say password incorrect". The cleaned text was then converted into TF-IDF features using unigrams and bigrams.

8. Issue Type Classification

Approach: TF-IDF features (313 terms) from the cleaned message were used to train a Random Forest Classifier (500 trees), with an 80/20 stratified split (80,000 train, 20,000 test).

Table 8.1: Issue type classification results

Metric	Value
Accuracy	100%
Precision / Recall / F1 (all 8 classes)	1.00

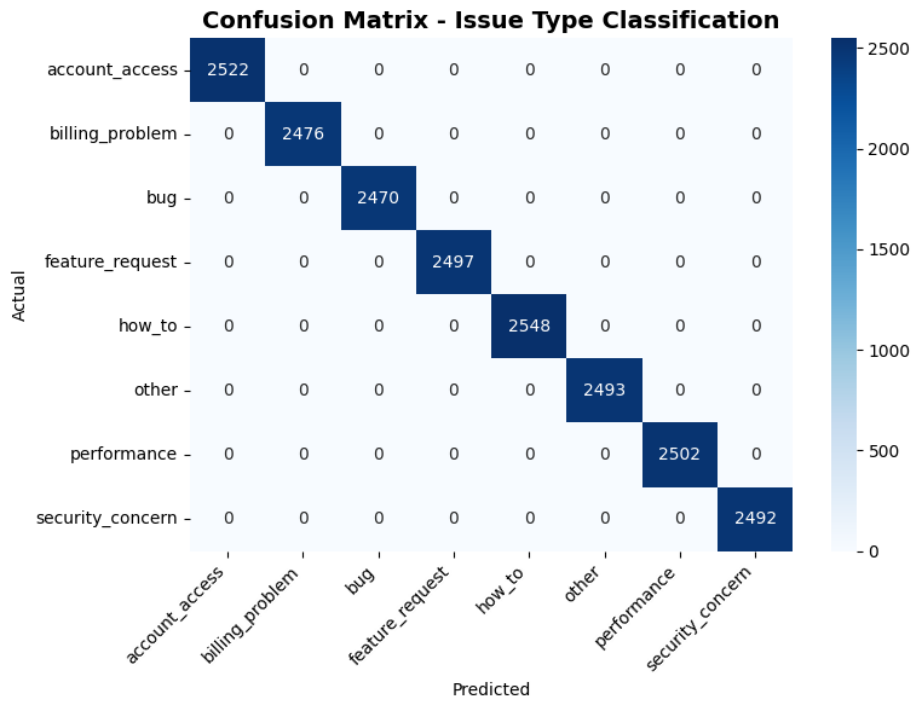


Figure 8.1: Confusion matrix – issue type classification

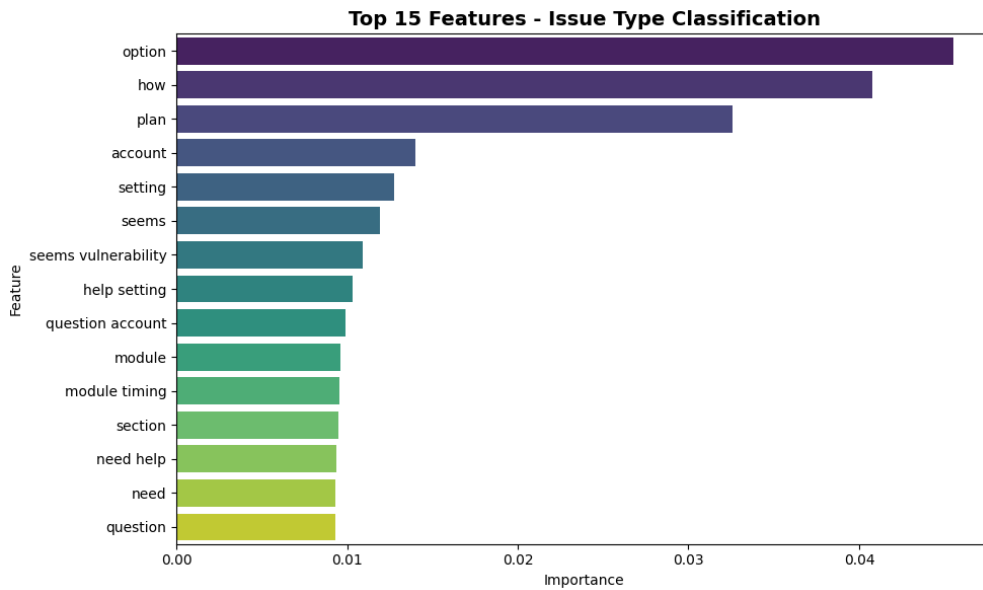


Figure 8.2: Top 15 TF-IDF features for issue type classification

Sanity check: The 100,000 messages come from only 96 unique templates, and each template maps to exactly one issue type, so a random split lets the model memorize templates. To test real generalization, a grouped split was used where test messages never appear in training. On unseen messages, TF-IDF + Logistic Regression reached **56.8% accuracy (Macro F1 = 0.629)** on a single grouped split. Because a single split contains only about 19 test messages, the evaluation was repeated with 5-fold GroupKFold, giving **Macro F1 = 0.605 ± 0.107**. This is the more realistic estimate of performance on new ticket wording.

Only the customer's message was used as input; agent replies and resolution summaries were excluded because they reveal the issue type directly (leakage).

9. Team Routing & BERT vs TF-IDF

9.1 Team Routing

Approach: Each ticket is routed to the responsible team, represented by `product_area` (7 teams). Models used the message text, optionally combined with metadata available at ticket creation (channel, platform, customer segment).

Table 9.1: Team routing results (random split, 20,000 test tickets)

Model	Accuracy	Macro F1
LogReg (text)	0.574	0.580
LogReg (text + meta)	0.573	0.575
LinearSVC (text + meta)	0.572	0.575
Random Forest (text + meta)	0.574	0.574
Baseline (most frequent)	0.144	0.036

- The best model is **4× better than the baseline**, and the correct team is among the top 3 suggestions for **71.6%** of tickets. A top-3 list is practical for agents, who can confirm the right team with one click.
- Metadata did not improve results, which is consistent with the EDA finding that these attributes are uniformly distributed.
- On unseen messages, a single grouped split gave a Macro F1 of 0.659, but 5-fold GroupKFold gave a lower and more reliable 0.516 ± 0.112 . The single split was a favorable draw, since each test set contains only about 19 unique messages.
- **Error analysis:** Most of the 8,518 errors come from generic messages (e.g., "I have a general question about my account" or "I noticed a suspicious login on my account") that appear across several teams, so the text alone cannot identify the team.

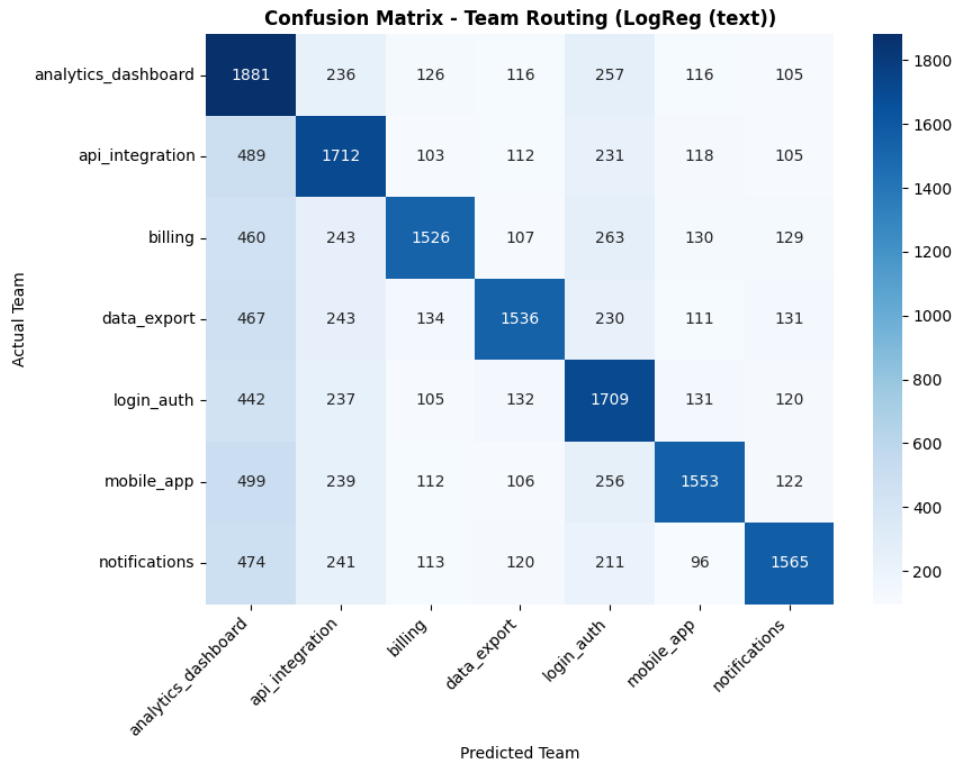


Figure 9.1: Confusion matrix – team routing (LogReg, text)

A `route_ticket()` function was built to return the top-3 teams with probabilities for any new ticket, ready for use in the dashboard.

9.2 BERT vs TF-IDF

Approach: To test the proposal's hypothesis, messages were embedded with a pre-trained Sentence-BERT model (all-MiniLM-L6-v2, 384 dimensions) and compared with TF-IDF, both using Logistic Regression. The model was not fine-tuned, because with only 96 unique messages fine-tuning would memorize the templates rather than learn language patterns.

Table 9.2: BERT vs TF-IDF (Macro F1)

Task	Split	TF-IDF	BERT
Issue type	Random	1.000	1.000
Issue type	Grouped (unseen)	0.629	0.629
Team routing	Random	0.590	0.580
Team routing	Grouped (unseen)	0.662	0.634

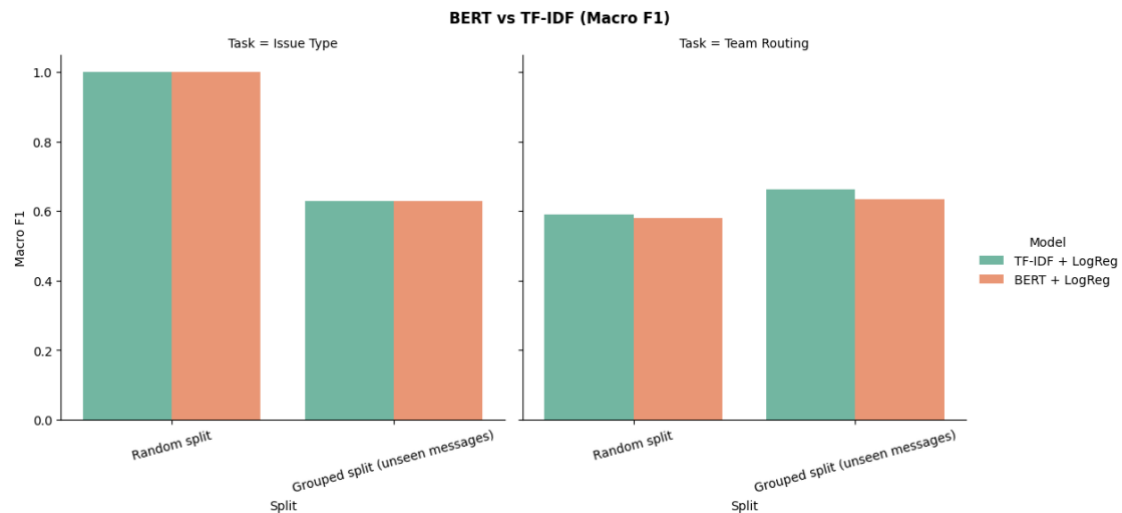


Figure 9.2: BERT vs TF-IDF comparison (Macro F1)

Finding: BERT performed on par with TF-IDF and slightly lower for team routing. The hypothesis that BERT would clearly outperform keyword-based methods was not supported on this dataset, because the templated messages are short and keyword-driven. BERT's contextual strengths are expected to matter more with real, varied user text.

Note: The grouped-split results in Table 9.2 come from a single split and may vary by several points. The 5-fold GroupKFold results in Sections 8 and 9.1 are the more reliable estimates.

10. Resolution Time Prediction

Approach: Models were trained only on resolved tickets (50,131), since resolution time is only meaningful for tickets that were actually resolved. A median baseline was used for comparison.

Leakage findings during development:

- An early model reached $R^2 = 0.58$. Investigation showed this came from status, which is only known after closure.
- Another version reached $R^2 = 1.00$ because a capped copy of the target column had been included as a feature.
- Both issues were fixed before the final evaluation.

Table 10.1: Resolution time model comparison (test set)

Model	MAE (hours)	RMSE (hours)	R^2
Baseline (median)	15.170	18.807	-0.051
Gradient Boosting (MAE loss)	15.177	18.809	-0.052
HistGradientBoosting (MAE loss)	15.177	18.786	-0.049
CatBoost (MAE loss)	15.207	18.816	-0.052
LightGBM (MAE loss)	15.262	18.857	-0.057
XGBoost (MAE loss)	15.269	18.501	-0.018
Random Forest (log target)	15.309	19.485	-0.129
Neural Network (MLP)	15.503	18.443	-0.011
Random Forest (default)	15.75	18.69	-0.04

Hyperparameter tuning of LightGBM (RandomizedSearchCV, 30 candidates, 5-fold CV) gave a test MAE of 15.166 hours, still equal to the baseline (15.170).

Why do all models match the baseline? Mutual information between each feature and resolution time was measured. All values were below 0.008 (the highest was channel_web_form at 0.0073), meaning the features carry almost no information about resolution time.

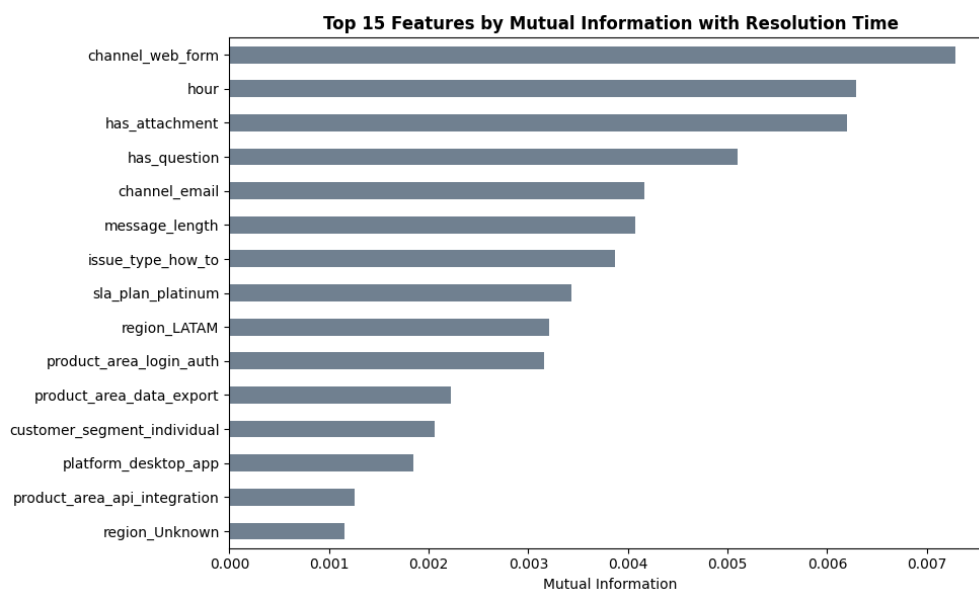


Figure 10.1: Top 15 features by mutual information with resolution time

Summary: Gradient Boosting with an absolute-error loss is the recommended model because it directly optimizes MAE and matches the baseline, but no model can beat the baseline with the available features. Reliable prediction would require real-world data with features such as team workload, agent experience, and ticket complexity.

11. Anomaly Detection

Approach: Tickets were aggregated per day (1,460 days) using three metrics: ticket count, average resolution time, and reopen rate. Daily aggregation was chosen over hourly because hourly volumes were too low (2.85 tickets per hour on average). Two unsupervised models were applied with a contamination rate of 5%:

- **Isolation Forest:** isolates extreme values.
- **Local Outlier Factor (LOF):** compares each day to its nearest neighbors (features were standardized first).

Table 11.1: Anomaly detection results

Metric	Value
Average daily tickets	68.5 (min 43, max 95)
Anomalous days (Isolation Forest)	73
Anomalous days (LOF)	73
Agreed by both models (high-confidence)	45
Injected surges detected (validation)	10 / 10

The top high-confidence anomalies had 85–95 tickets per day, 25–40% above the daily average; the highest was 2025-03-29 with 95 tickets. Some anomalies occur at normal ticket volumes because they are driven by unusual resolution times or reopen rates.

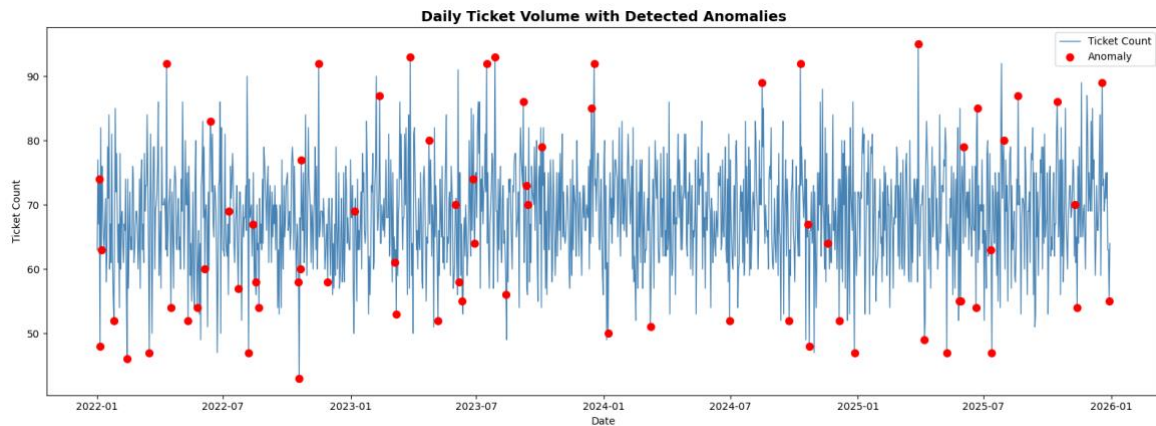


Figure 11.1: Daily ticket volume with detected anomalies

The same anomalies are shown on the daily average resolution time below. Several flagged days have normal ticket volumes but unusually high or low average resolution times, which confirms that the models use all three metrics together rather than ticket count alone.

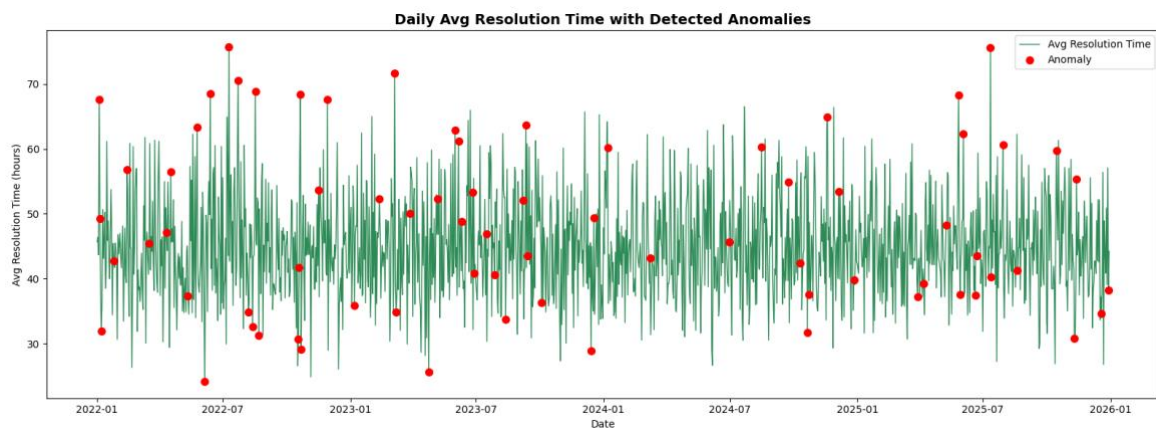


Figure 11.2: Daily average resolution time with detected anomalies

Validation: Since the data has no labelled incidents, the detector was tested by doubling the ticket count on 10 random days. Isolation Forest detected all 10 injected surges, showing it reliably catches real volume spikes.

11.1 Anomalies by Issue Type

Daily counts per issue type were analyzed with a second Isolation Forest. For each anomalous day, the issue type with the largest spike relative to its own average (z-score) was identified. Spikes were spread across all issue types (6–12 days each), led by other, billing_problem, and how_to.

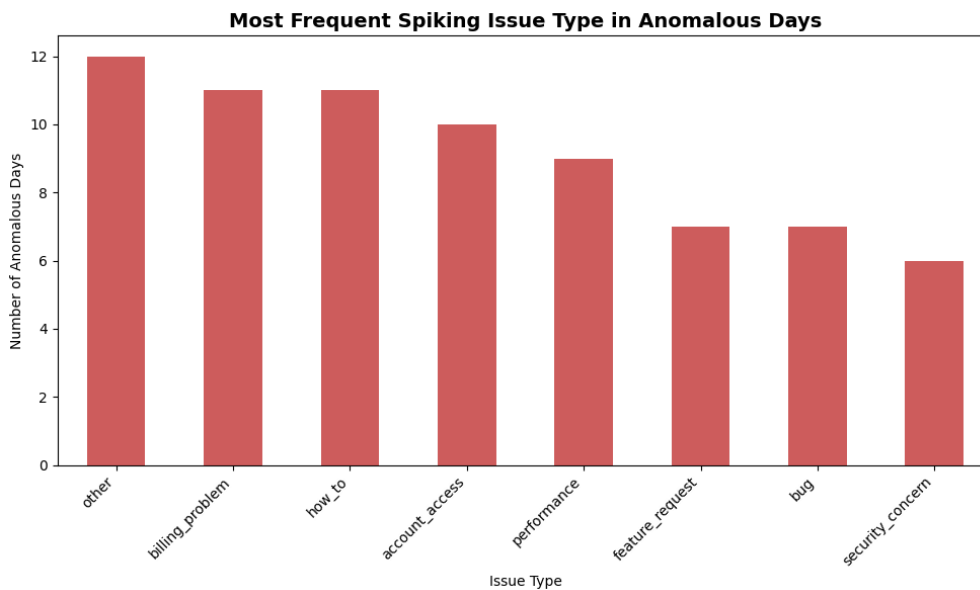


Figure 11.3: Most frequent spiking issue type in anomalous days

- A chi-square test gave $p = 0.80$, so the differences between issue types are not statistically significant; no single issue type drives the anomalies.
- 12 days had strong spikes ($z \geq 3$), such as feature_request on 2024-07-07 ($z = 3.92$) and other on 2022-01-29 ($z = 3.91$). These are the days most worth alerting on.

Note: The 5% contamination rate forces the models to flag a fixed share of days. In production, this threshold would be set with input from the support team.

12. Interactive Dashboard

An interactive dashboard was built with Streamlit and Plotly to make the results usable by helpdesk managers. It loads the saved models (TF-IDF vectorizer, issue type classifier, label encoder, and team routing model) and the exported result files, so it runs without retraining. It has four pages, selected from the sidebar.

12.1 Service Desk Overview

The overview page shows key performance indicators: 100,000 total tickets, 39,887 unresolved tickets, a median resolution time of 29.9 hours, an average CSAT of 3.20 out of 5, and a 5.0% reopen rate. Interactive charts show tickets by team, by issue type, and by hour of day, the status breakdown, and weekly ticket volume. A date range filter in the sidebar updates all charts.



Figure 12.1: Dashboard – Service Desk Overview

12.2 Ticket Classifier & Routing

Managers can type a new ticket message (or pick an example) and select the channel, platform, and customer segment. The dashboard predicts the issue type with a confidence score and suggests the top 3 teams with their probabilities, using the same preprocessing and models from Sections 8 and 9.

Deploy

Classify & Route a New Ticket

Type a ticket message and the system predicts the **issue type** and the **team** it should go to.

Pick an example message (optional)

(write your own)

Ticket message

how can i export my data from billing section?

Channel: chat | Platform: ios | Customer segment: individual

Classify ticket

Issue Type

Predicted

how_to

↑ 100% confidence

Suggested Teams

- 1. billing — 100%**
- 2. data_export — 0%**
- 3. notifications — 0%**

Figure 12.2: Dashboard – Classify and route a new ticket

12.3 Anomaly Monitor

The anomaly page lets the user choose the detection method (Isolation Forest, LOF, or both agreeing for high confidence) and the metric to plot. It displays the number of anomalous days, the timeline with anomalies highlighted, and a table of anomalous days. With both models agreeing, 45 days (3.1%) are flagged, compared with 73 days (5.0%) for Isolation Forest alone.

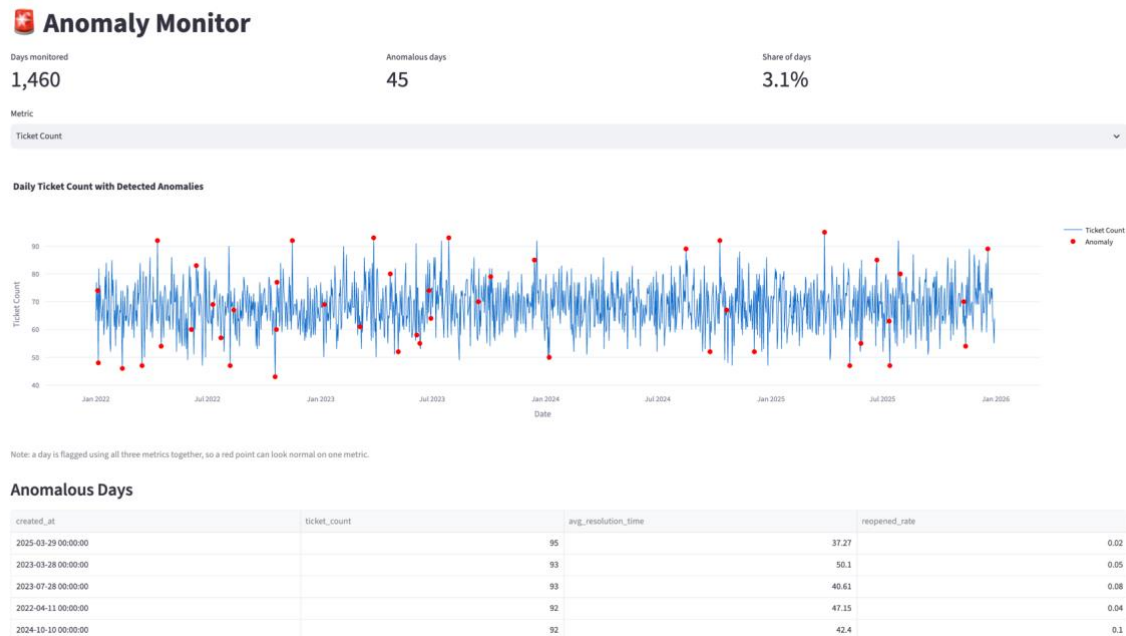


Figure 12.3: Dashboard – Anomaly Monitor (high-confidence anomalies)

A second section shows which issue type spiked on each anomalous day. A slider sets the minimum spike strength (z-score); at the recommended threshold of $z \geq 3$, 12 days are listed as strong spikes that should trigger alerts.

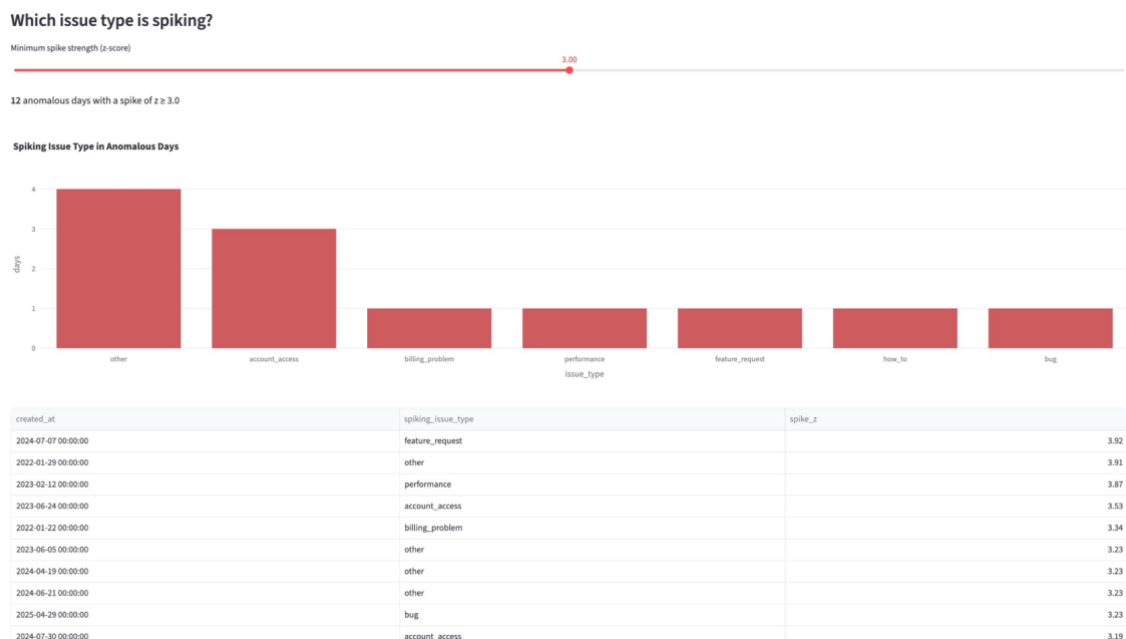


Figure 12.4: Dashboard – Spiking issue types with z-score threshold

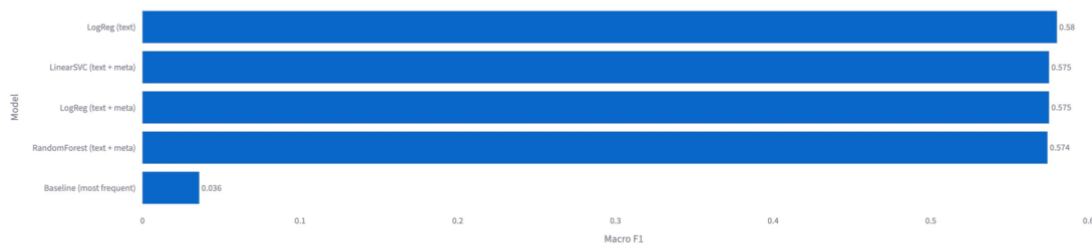
12.4 Model Performance

The final page compares the team routing models, summarizes the key findings of the project, and provides tabs with the main evaluation charts (BERT vs TF-IDF, confusion matrices, mutual information, and anomaly spikes) presented in Sections 8–11.

Model Performance

Team Routing — Model Comparison

Macro F1 by Model



Model	Accuracy	Macro F1
LogReg (text)	0.574	0.58
LogReg (text + meta)	0.573	0.575
LinearSVC (text + meta)	0.572	0.575
RandomForest (text + meta)	0.574	0.574
Baseline (most frequent)	0.144	0.036

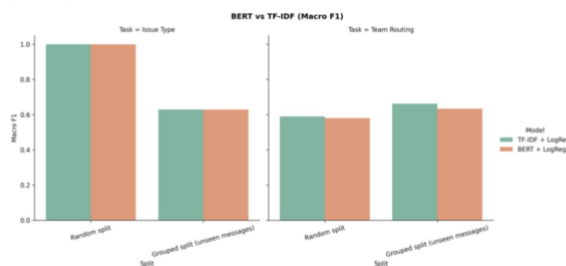
Figure 12.5: Dashboard – Team routing model comparison

Key Findings

- Issue type classification reaches 100% on known message templates. The dataset has only 96 unique messages, so we also tested on unseen messages (5-fold GroupKFold): **Macro F1 = 0.605 ± 0.107** , which is the realistic estimate for new wording.
- Team routing is limited by the data: the same message can belong to different teams. On unseen messages, **Macro F1 = 0.516 ± 0.112** , so the top-3 suggestions are more useful than a single prediction.
- Resolution time could not be predicted better than the median baseline: none of the available features carries information about it (confirmed with mutual information).
- BERT vs TF-IDF: pre-trained BERT embeddings did not outperform TF-IDF on this dataset, because the messages are short templates where keywords already carry the meaning.
- Anomaly detection flags unusual days; only strong spikes ($z \geq 3$) should trigger alerts.

Evaluation Charts

BERT vs TF-IDF Routing Confusion Matrix Issue Type Confusion Matrix Resolution Time (MI) Anomaly Spikes



Note: grouped-split results here come from a single split and may vary by several points. The 5-fold GroupKFold results in Key Findings are more reliable.

Figure 12.6: Dashboard – Key findings and evaluation charts

Running the dashboard: The app runs locally with `streamlit run app.py`. The scikit-learn version is pinned to 1.6.1, matching the version used to save the models in Colab, to avoid compatibility errors when loading them.

13. Results Summary

Table 13.1: Summary of results

Task	Best model	Key result
Issue type classification	TF-IDF + Random Forest	100% (random split); Macro F1 0.605 $\pm$ 0.107 on unseen messages
Team routing	TF-IDF + Logistic Regression	57.4% accuracy, 71.6% top-3 (baseline 14.4%); Macro F1 0.516 $\pm$ 0.112 on unseen messages
BERT vs TF-IDF	—	BERT on par with TF-IDF
Resolution time	Gradient Boosting (MAE loss)	MAE $\approx$ 15.2 hours, at baseline level
Anomaly detection	Isolation Forest + LOF	45 high-confidence days; 10/10 injected surges detected
Dashboard	Streamlit + Plotly	4 interactive pages for managers

14. Limitations

- **Synthetic data:** Distributions are largely uniform, and relationships found in real support data (e.g., issue type affecting resolution time) are absent.
- **Templated text:** Only 96 unique customer messages exist, which makes issue classification trivial on a random split and limits the value of advanced NLP such as BERT. It also makes results on unseen messages unstable (about ± 0.11 across folds).
- **No priority labels:** Priority assignment could not be implemented.
- **Generic messages:** Some messages do not indicate a specific team, which caps routing accuracy.
- **Fixed contamination rate:** Anomaly counts depend on the chosen 5% threshold, and there are no labelled incidents to measure precision.

15. Next Steps

- Test the pipeline on real ticket data with varied language.
- Fine-tune BERT once enough diverse text is available.
- Add workload and agent features to enable meaningful resolution time prediction.
- Deploy the dashboard to a shared server (e.g., Streamlit Community Cloud) and connect it to live ticket data.

16. Conclusion

This project delivered an end-to-end IT helpdesk analytics pipeline covering data cleaning, EDA, NLP-based issue classification, team routing, resolution time prediction, and anomaly detection. Beyond the headline numbers, careful validation was a key outcome: grouped splits revealed template memorization in classification, leakage checks prevented misleading resolution time results, and injected surges verified the anomaly detector.

Team routing and anomaly detection provide the most practical value, and the Streamlit dashboard puts them in the hands of helpdesk managers: routing narrows the correct team to a top-3 list for 71.6% of tickets, and the anomaly detector reliably flags unusual days for investigation. The main

limitation is the synthetic dataset, and applying the same pipeline to real ticket data is the most important next step.

References

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- NLTK – <https://www.nltk.org/>
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